STAGE 2 – PRACTICAL ENGINEERING CHALLENGE

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Problem Statement: Industrial Manufacturing Process Automations

Industrial manufacturing depends on highly controlled liquid-handling systems to deliver process fluids reliably from a source or bulk supply to downstream process equipment. Depending on the application, these fluids may include **ultrapure fluids, process chemicals, solvents, developers, or CMP-related fluids**. Industry standards explicitly address liquid-chemical distribution, high-purity piping, bulk chemical distribution, system qualification, and UPW system operation and monitoring.

In such systems, **flow is a fundamental process variable**. An incorrect flow rate can indicate or cause a restriction, incorrect valve state, inadequate pumping performance, abnormal process demand, or another hydraulic deviation. The consequence may be unstable process conditions, incorrect chemical delivery, equipment malfunction, or loss of process performance.

This challenge therefore focuses on a fundamental industrial facilities and process-engineering task: **To establish a defined process-fluid flow, measure it reliably, detect a deviation, diagnose its cause, and restore the process to the required operating condition.**

Engineering Challenge

As you are part of a multidisciplinary engineering team responsible for designing, building and validating a small-scale automated liquid-flow system represented of industrial process handling system.

CASE 1 – CONTROL + MEASUREMNT

Problem: Develop a system that manages the changing level of fluid in the sump automatically (when there is a sudden external fluid charging to the sump) and while accurately measuring and totalizing the fluid discharged from the sump using the flow transmitter.

CASE 2 – FLUSHING +SHUTDOWN

Problem: Develop an alternative operating configuration that allows flush water to pass through the system while all three pumps operate simultaneously, and then safely transition the system to an OFF condition when the flush-water supply has been exhausted.

Given Process Parameters

Pipe diameter =	**8mm**
Fluid density (ρ) =	**1000kg/m^3**
Dynamic Viscosity (μ) =	**1x10^{-3}Pa.s**
Required Flow range =	**1-6 L/min**
Nominal target flow for the live challenge =	**3L/min**

Available Hardware

Mechanical Hardware	Electronic Hardware
1. Water Tank – 3 Quantity	1. Custom PCB
2. Pumps – 3 Quantity	2. ESP32 NodeMCU module
3. FT-01 Flow Transmitter – 01 Quantity	3. Pump Driver-1
4. Ball valve – 3 Quantity	4. Pump Driver-2
5. ON/OFF/XV Valve – 1 Quantity	5. Pump Driver-3
6. 10-mm OD tubing/blue – 9 meters/1 meter	6. 12 V DC power input (**SMPS**)
7. Microcontroller/PCB/ data-acquisition software	7. 12 VDC Solenoid-valve interface
8. Required electrical/ electronic accessories	8. Flow-sensor interface
	9. Float-switch interface
	10. ON/OFF-switch interface
	11. Pump-driver PWM/enable interfaces
	12. On/Off Switch

The team must determine the appropriate arrangement, connections and measurement method.

- Assemble and commission the liquid-flow system using the components provided.
- Measure and characterize the flow using FT-01
- Interface FT-01 with the provided **data-acquisition system**
- Experimentally determine the relationship between sensor output and actual flow.
- Characterize the effect of valve position on flow rate. Record and analyse flow vs time data
- Maintain target flow of 3L/min
- Use graphs and measured data to justify engineering decisions
- Design an appropriate hydraulic arrangement and P&ID

TEAM INSTRUCTIONS

- As a team go through the list provided to read about each component that has been Listed.
- The PCB is a 12 V DC and the SMPS connections should be made carefully.

TASK 1: HYDRAULICS SYSTEM DESIGN

Q1.1 Design and build the hydraulic path with the flow transmitter on the recirculation line. Submit/sketch a labelled P&ID below by showing the pumps, Needle valves, flow transmitter, flow direction, Tank, sump, tube diameter and measurement points.

CASE 1 – Process Fluid Transfer & Automatic Level Protection

Objective: Set-up and operate a closed-loop process- fluid handling system using the available tank, pump, valves flow transmitter, sump, float sensor and control PCB. The System must continuously transfer process fluid recirculation system by connecting the 10L fluid tank with a On/Off valve to a pump-01 and the pump-01 goes to the inlet of sump 4.2L tank. The Sump 4.2L outlet line is connected to another pump-02 followed by a flow transmitter that measures the total flow back to the 10L fluid tank. The Pump-01 runs on 50% duty cycle and Pump-02 runs on 100% duty cycle. **Calibrate the flow transmitter** and demonstrate correct fluid circulation, if there is sudden introduction of external fluid into the Sump tank then Pump-01 should stop while pump – 02 should still runs till fluid level reduces in the sump. After the fluid reduction in the SUMP 4.2L tank the Pump-01 should turn on and the process must continue. **Also, demonstrate Flow vs Time trends in mL/s, total transferred Liters back to the 10L tank** (share Screenshots flow in L/min). **Share the code.** The case-1 can be turned on using the available on/off Switch, to start the system.

CASE 2 – Flush/Line Conditioning Sequence

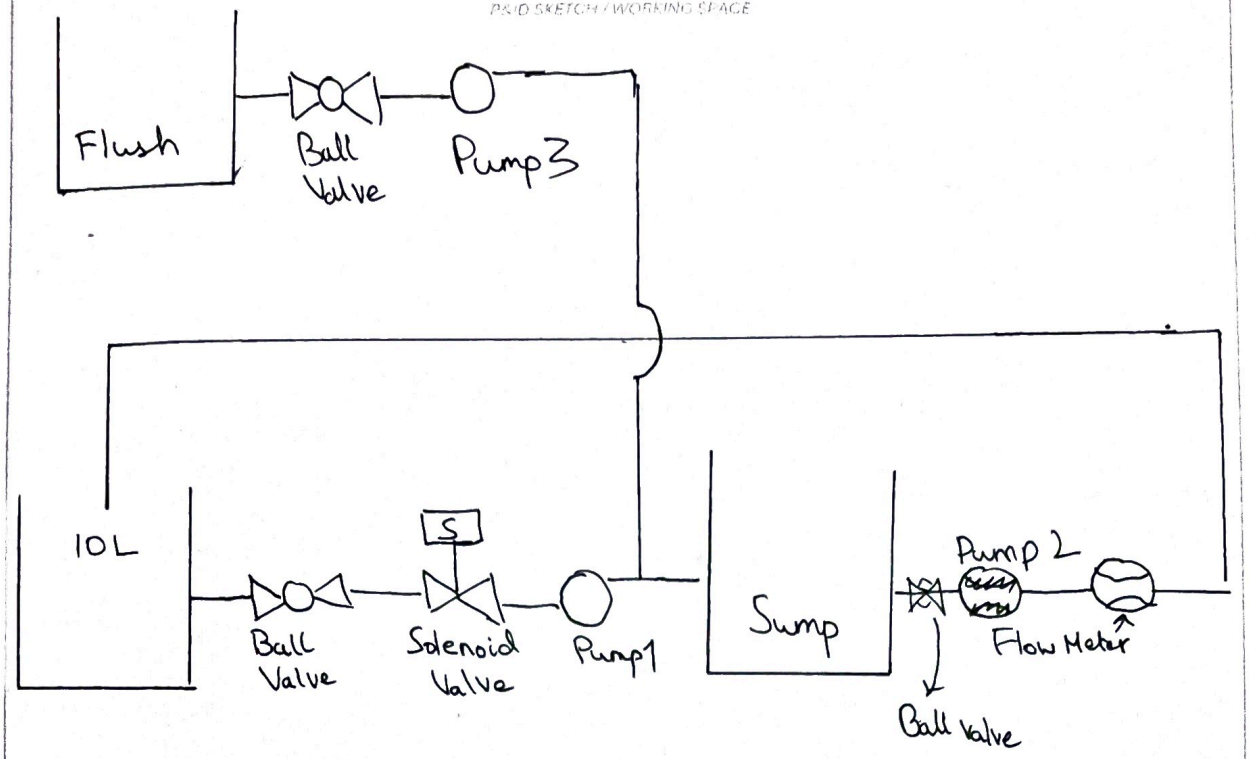
Objective: After successfully completing case 1, initiate the flush water in the sequence using the **equal Tee.** The On/Off valve must be switched off in order to isolate the 10L process fluid tank. The flush Fluid tank (4.2L) is used as a temporary fluid source to flush out the process fluid or any debris from the system. Before the start of this process make sure you open the manual ball valve on the fluid flush line before Pump-03 starts. The entire process of case -1 and case - 2 should run in a single go without manual intervention except opening of ball valve before the Pump-03 starts. **The System must stop as soon as the tank flush fluid 4.2L tank finishes.** The system must be automated without human interventions (only ball valve can be opened manually before the start of flush line sequence)

SUBMIT:

A. Submit: 10-15 min Video and Photograph of the completed setup, with identification of each component and explain the direction of flow in the video verbally. Demonstrate both Case - 1 and Case – 2.

B. Final P&ID Block diagram

P&ID SKETCH / WORKING SPACE



Q1.2 After setting-up the system, now operate it at approximately at 1L/min, 3L/min & 6L/min. Compare the theoretical velocity with the actual operating behaviour.

WORKING / COMPARISON

Theoretical vs Actual:

Flow = {1L/m, 3L/m, 6L/m}

$\Rightarrow V_{theory}$ = Flow/Area = {0.849 m/s, 2.546 m/s, 5.093 m/s}

$\vec{V}_{actual}$ = {0.781 m/s, 2.292 m/s, 4.380 m/s}

Reynolds Number Calc:

$Re = \rho \cdot V_{theory} \cdot d/\mu \Rightarrow Re$ = {4244, 12732, 25465}

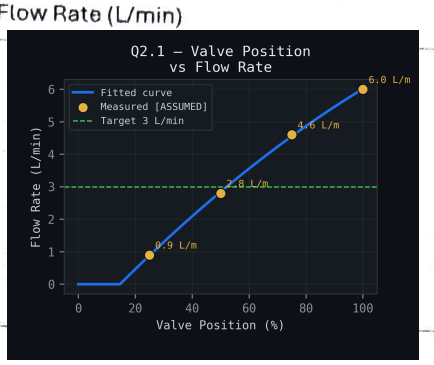
It is all turbulent flow.



TASK 2: FLOW RESPONSE TO VALVE POSITION

Q 2.1 Set the ball valve to approximately 25%,50%,75% & 100%. At each position, allow the flow to stabilize and record the flow rate. Create the following graph of valve position (%) vs Flow Rate (L/min)

VALVE POSITION	FLOW RATE
25%	0.9L/m
50%	2.8L/m
70%	4.6L/m
100%	6.0L/m



Q2.2 Determine the experimental calibration equation: $Q = af + b$, where Q = flow rate in L/min, f = pulse frequency in Hz, a = experimental calibration coefficient and b = offset.

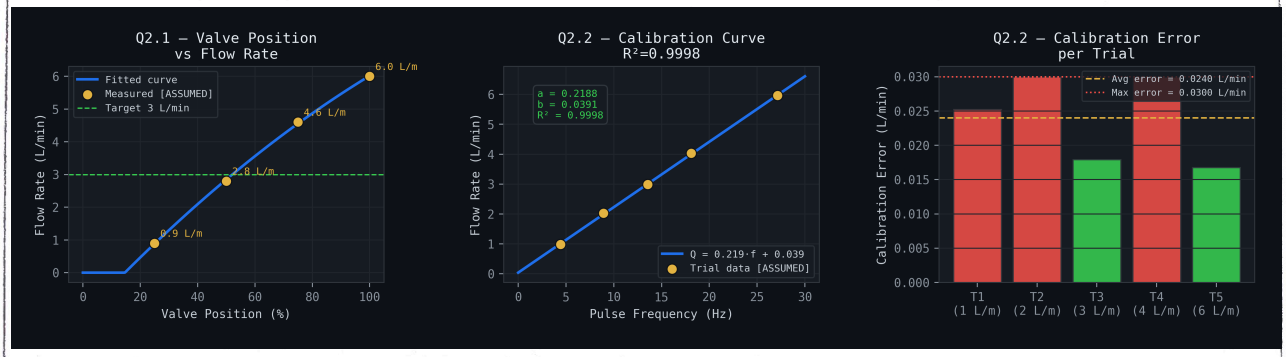
Report:

R^2 =	0.9998
a =	0.2188
b =	0.0391
Maximum calibration error =	0.03 L/min
Average calibration error =	0.024 L/min

Q2.3 FT-01 produces pluses corresponding to water flow. Run the system at least for 5 different flow conditions within the available operating range (Hint: use formulas f=N/t and Q=V/t). For each condition, record the following:

Trial	Pulses	Time(s)	Frequency (Hz)	Collected Volume (L)	Actual Flow (L/min)
1	133	30	4.43	0.492	0.984
2	268	30	8.93	1.012	2.024
3	407	30	13.56	1.495	2.99
4	543	30	18.1	2.015	4.03
5	814	30	27.13	2.98	5.96

CALCULATIONS / OBSERVATIONS



TASK 3: HYDRAULIC SYSTEM THEORY AND FAULT ANALYSIS

Q3.1 The effective liquid volume inside a section of tubing is 0.5L, while the operating flow rate is 2.5L/min. The hydraulic residence time is given by $T_r = Q/V$. Calculate the residence time in seconds, also answer the following questions:

1. What does residence time physically represent in this system?
2. What happens to residence time if the tubing volume is doubled while the flow rate remains constant?
3. What happens to residence time if the flow rate is doubled while the tubing volume remains constant?
4. Why is residence time important when interpreting a sudden change in measured flow?

ANSWER

Residence time is given by $T_r = V/Q$ not Q/V

So, $T_r = V/Q = 0.5L / 2.5L/min = \boxed{12 \text{ seconds}}$

(1) Residence time represents the average time a fluid element spends inside the tubing section from entering to leaving, or it is the time required to replace the tube's contents with fresh fluid

(2) T_r doubles, when $V = 1L$, $Q = 2.5L/min$

$\Rightarrow T_r = 24s$

(3) T_r halves, when $Q = 5L/min$, $V = 0.5L$, $T_r = 6sec$

(4) It is important because, residence time sets the delay and smoothing between what happens at one point in the line and what a sensor sees elsewhere &. A disturbance will take roughly T_r amount of time to travel through a section, so a sensor downstream will not show change until 12 s later. Controllers and measurement intervals need to account for this lag.

Q3.2 The following observation are obtained from the same system:

CONDITION	PUMP	VALVE	FT-01 PULSE SIGNAL	MEASURED FLOW
A	ON	Open	Normal	Normal
B	ON	Partially closed	Reduced	Reduced
C	ON	Open	Zero	Zero
D	OFF	Open	Zero	Zero
E	ON	Open	Intermittent	Fluctuating

For each of the condition, identify the most likely physical or instrumentation condition. Then answer: Which condition is most difficult to distinguish using FT-01 alone, and what additional information would you require to distinguish the fault? (Hint: Teams must consider possibilities such as: Valve restriction, pump failure, hydraulic blockage, sensor/signal failure, unstable flow etc)

ANSWER → Normal operation, pump is running and the path is open, both the flow reading and the pulse agree

B → Valve restriction, pump is running and valve is partially closed, so the flow is lower. The pulse rate falls proportionally to the flow, which means the sensor is reporting a real change.

C → Hydraulic Blockage or sensor/signal failure

D → Pump failure

E → Unstable flow

Condition C is hardest to seperate because 'no flow' and 'no signal' look identical at FT-01, as real blockage and a dead sensor both give zero pulses and zero flow.

Additional information needed would be pump discharge pressure, pump motor current, a second flow meter to independently check whether liquid is flowing.

```
// Define GPIO pins


const int MOTOR1[4] = {18, 23, 17, 16};
const int MOTOR2[4] = {33, 32, 26, 13};
const int MOTOR3[4] = {25, 23, 22, 21};


// PWM Configuration Parameters
const uint32_t PWM_FREQ = 5000; // 5 kHz frequency (typical for DC motors)
const uint8_t PWM_RES   = 8;    // 8-bit resolution (0 - 255)

const int FLOAT_SWITCH = 35;


const byte SENSOR_PIN = 14;
const int SOL_VALVE = 27;
const int BUTTON = 34;

volatile unsigned long pulseCount = 0;

unsigned long lastTime = 0;
float flowRate = 0.0;
float totalLitres = 0.0;

// Common YF-S401 calibration factor: approximately 5880 pulses/litre.
// Verify this with your specific sensor.
const float PULSES_PER_LITRE = 5880.0;

void IRAM_ATTR pulseCounter() {
  pulseCount++;
}

void setup() {
  // Configure static direction pins
  ledcAttach(MOTOR1[0], PWM_FREQ, PWM_RES);
  ledcAttach(MOTOR2[0], PWM_FREQ, PWM_RES);
  ledcAttach(MOTOR3[0], PWM_FREQ, PWM_RES);

  pinMode(MOTOR1[1], OUTPUT);
  pinMode(MOTOR1[2], OUTPUT);
  pinMode(MOTOR1[3], OUTPUT);
```

```
  pinMode(MOTOR2[1], OUTPUT);
  pinMode(MOTOR2[2], OUTPUT);
  pinMode(MOTOR2[3], OUTPUT);
  pinMode(MOTOR3[1], OUTPUT);
  pinMode(MOTOR3[2], OUTPUT);
  pinMode(MOTOR3[3], OUTPUT);

  pinMode(FLOAT_SWITCH, INPUT);
  pinMode(BUTTON, INPUT);
  pinMode(SOL_VALVE, OUTPUT);
  digitalWrite(SOL_VALVE, LOW);


  digitalWrite(MOTOR1[1], LOW);
  digitalWrite(MOTOR1[2], HIGH);
  digitalWrite(MOTOR1[3], HIGH);

  digitalWrite(MOTOR2[1], LOW);
  digitalWrite(MOTOR2[2], HIGH);
  digitalWrite(MOTOR2[3], HIGH);

  digitalWrite(MOTOR3[1], LOW);
  digitalWrite(MOTOR3[2], HIGH);
  digitalWrite(MOTOR3[3], HIGH);


  Serial.begin(115200);

  pinMode(SENSOR_PIN, INPUT_PULLUP);

  attachInterrupt(
    digitalPinToInterrupt(SENSOR_PIN),
    pulseCounter,
    FALLING
  );

  Serial.println("YF-S401 Sensor Test");
  Serial.println("Waiting for pulses...");
  lastTime = millis();
}

int TOGGLE = 0;

void loop() {

  unsigned long currentTime = millis();

  if (currentTime - lastTime >= 1000) {
    noInterrupts();
    unsigned long pulses = pulseCount;
    pulseCount = 0;
```

```
  interrupts();

  // Pulses counted in the last one-second interval
  flowRate = (pulses * 60.0) / PULSES_PER_LITRE;

  // Convert pulses to litres and accumulate
  totalLitres += pulses / PULSES_PER_LITRE;

  Serial.print("Pulses/sec: ");
  Serial.print(pulses);

  Serial.print(" | Flow rate: ");
  Serial.print(flowRate, 3);
  Serial.print(" L/min");

  Serial.print(" | Total: ");
  Serial.print(totalLitres, 4);
  Serial.println(" L");

  lastTime = currentTime;
}

int button=digitalRead(BUTTON);
if(!TOGGLE && !button) TOGGLE = 1;
if(!TOGGLE){
    digitalWrite(SOL_VALVE, HIGH);
    int fs = digitalRead(FLOAT_SWITCH);
    if(!fs){
      ledcWrite(MOTOR1[0], 0);
      ledcWrite(MOTOR2[0], 255);
    }
    else{
      ledcWrite(MOTOR1[0], 125);
      ledcWrite(MOTOR2[0], 255);
    }
}

else{
  digitalWrite(SOL_VALVE, LOW);
  ledcWrite(MOTOR1[0], 0);
  ledcWrite(MOTOR2[0], 255);
  ledcWrite(MOTOR3[0], 255);
}



}
```